

Adaptive Radar Detection of Dim Targets Assisted by RIS

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Abstract—We present and assess two detection architectures for low observable targets that exploit the backscattered echoes from a reconfigurable intelligent surface (RIS). This approach represents a low-cost alternative to the use of a multistatic radar network with its synchronization, costs, phase coherence, and energy consumption issues. Positioning the RIS in the region of interest leads to joint monostatic and bistatic configurations that collect the energy backscattered by the stealth target along irrelevant directions thereby increasing the target energy at the receiver. The analysis is conducted in comparison with classical detectors and in terms of probability of detection. Moreover, we provide some important guidelines for the RIS design and tapering in order to come up with an effective system configuration. The results point out that this new approach can reveal low observable targets unlike the conventional one.

Index Terms—Detection, likelihood ratio test, low-RCS, radar, radar cross section, reflective intelligent surfaces.

I. INTRODUCTION

In the last years, reconfigurable intelligent surfaces (RISs) have become a key technology for next-generation wireless communications [1], [2, and references therein]. These surfaces can be broadly classified as passive Smart Electromagnetic Skins or reflectarrays, which provide fixed beamforming, and fully reconfigurable versions capable of dynamic wave control [3]. Their real-time programmability is enabled by semiconductor elements such as PIN and varactor diodes [4], enabling manipulation of phase, amplitude, and polarization [5]. Advanced architectures integrate sensing functions for joint communication and sensing [6], or active components to compensate for double-path loss [7]. This hardware versatility positions RIS as a cornerstone for smart, programmable radio environments in 5G-Advanced and 6G networks.

Although the initial application of RISs has been in the context of communication networks, their extension to sensing systems is rapidly progressing within the integrated sensing and communication (ISAC) paradigm [8]. An endless list of

contributions examine coexistence, cooperation, and codesign between radar and communication systems. For instance, [9] proposes a RIS-assisted framework optimizing the signal-to-interference-plus-noise ratio (SINR) for joint radar detection and communication quality-of-service. Other works mitigate mutual interference [10], [11], or employ hybrid RISs for dual-function sensing-communication operations [12]. In 5G/6G scenarios, RISs are used to detect small non-cooperative targets via OFDM-based ISAC systems and track-before-detect techniques [13]–[15], while polarization-modulating RISs are shown to enhance target classification in distributed multiple-input multiple-output ISAC systems [16].

This paper focuses on RIS-aided detection architectures for airborne low radar cross section (RCS) targets whose low visibility is intentionally designed rather than size-induced as in [13]. Stealth or “low observable” technology has become central in modern airborne platforms [17], where shaping techniques reduce backscattered energy in critical angular sectors and redirect most of it toward *irrelevant directions*. Multistatic radars can counter this by exploiting spatial diversity, but they require costly and complex synchronization. Here, we propose that a monostatic radar can emulate multistatic benefits by exploiting suitably deployed RISs that intercept and redirect the energy scattered along the irrelevant directions.

Crucially, the electromagnetic surface considered in this work is hypothesized as a passive smart electromagnetic skin and its backscattering pattern is statically designed to illuminate all the radar cells under investigation simultaneously with a single, fixed beam. This is achieved through an adequate, pre-defined tapering of the surface’s unit cells to control the phase of the reflected wavefront. We analyze the geometric constraints of a search-radar scenario where both the target and the RIS are illuminated, and we formulate the associated detection problem as a binary hypothesis test. To solve it, we adopt: (i) an estimate-and-plug generalized likelihood ratio test (GLRT) framework [18]–[20] and (ii) an approximate GLRT based on upper-bound minimization. These approaches yield two different detectors that can capitalize all the available energy. Simulations on synthetic data demonstrate substantial

This work was supported in part by the National Natural Science Foundation of China under Grant No.62571525 and Grant No.62201564. The work of D. Orlando was partially supported by the Italian Ministry of Education and Research (MUR) in the framework of the FoReLab project (Departments of Excellence).

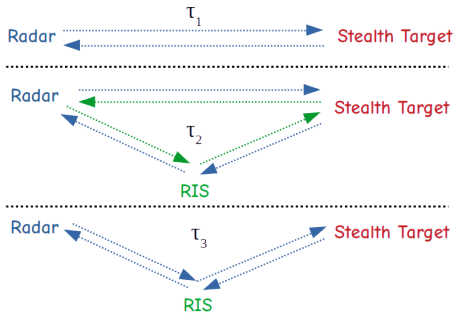


Fig. 1: Considered paths.

detection improvements compared to conventional methods. In summary, the fundamental novelty of this paper lies in addressing the detection of low-observable (stealth) targets with extremely weak radar returns by jointly exploiting echoes from different paths (radar-target-radar, radar-target-RIS-radar, and radar-RIS-target-RIS-radar within an adaptive detection framework for multiple point-like targets.¹

II. PROBLEM FORMULATION

Let us consider a monostatic radar system that uses a planar array of N_a tiles to sense the region of interest. The system transmits a coherent burst of N_p pulses that are Doppler tolerant [21] and the illumination is accomplished by means of a fan beam. A RIS is placed in the region of interest with the objective of illuminating prospective stealth targets from below and denote by $d_{RS} > 0$ the RIS-radar distance, $d_{RT} > 0$ the distance from the radar to the target, and $d_{ST} > 0$ the distance between the target and the RIS. The RIS configuration is such that it can manage the path radar-RIS-target and target-RIS-radar (see Fig. 1), where the path radar-target-radar (RTR) is the conventional monostatic radar direct echo, the radar-target-surface-radar (RTSR or RSTR) is the single-bounce RIS-assisted path, and the radar-surface-target-surface-rada (RSTSR) is the double-bounce RIS-assisted path. These three paths correspond to different propagation delays and thus fall into different range bins. Specifically, $\tau_1 = \frac{2d_{RT}}{c}$, $\tau_2 = \frac{d_{RT}+d_{RS}+d_{ST}}{c}$, and $\tau_3 = \frac{2(d_{RS}+d_{ST})}{c}$ (c is the speed of light) are the delays associated with the path RTR, RTSR, and RSTSR, respectively. Moreover, we assume that the geometry is such that

$$d_{RS} + d_{ST} - d_{RT} \geq 2\Delta r, \quad (1)$$

¹Symbols $\det(\cdot)$, $\text{Tr}(\cdot)$, $(\cdot)^T$, and $(\cdot)^\dagger$ denote the determinant, trace, transpose, and conjugate transpose, respectively. As to the numerical sets, $\mathbb{C}$ is the set of complex numbers and $\mathbb{C}^{N \times M}$ is the Euclidean space of $(N \times M)$ -dimensional complex matrices (or vectors if $M = 1$). For two sets A and B , the difference set is $A \setminus B$. Function $\text{rect}(x)$ is the rectangular window and is equal to 1 if $|x| < 1/2$ and to 0 if $|x| > 1/2$. $\mathbf{I}$ and $\mathbf{0}$ stand for the identity matrix and the null vector or matrix of proper size. Given $\mathbf{a} \in \mathbb{C}^{N \times 1}$, $\text{diag}(\mathbf{a}) \in \mathbb{C}^{N \times N}$ indicates a diagonal matrix whose i th diagonal element is the i th entry of $\mathbf{a}$. If $\mathbf{A} \in \mathbb{C}^{N \times N}$, $\text{diag}(\mathbf{A}) \in \mathbb{C}^{N \times N}$ is a diagonal matrix whose main diagonal is the same as $\mathbf{A}$. Symbol $\|\cdot\|_F$ is the Frobenius norm of the matrix argument. The acronym PDF stands for Probability Density Function. Finally, we write $\mathbf{x} \sim \mathcal{CN}_N(\mathbf{m}, \mathbf{M})$ if $\mathbf{x}$ is a complex circular N -dimensional normal vector with mean $\mathbf{m}$ and positive definite covariance matrix $\mathbf{M}$.

where Δr is the range resolution. This condition is primarily introduced to simplify the derivation and highlight the core concept; it can be met by suitably placing the RIS in the region of interest (an example is provided in Section IV). In addition, in scenarios where the condition is violated, the alternative detection strategies (e.g., adopting higher-resolution waveforms in conjunction with the proposed detection processing) are required.

Under the above assumptions, we jointly process a set of K_P consecutive range bins (to capitalize all the available energy unlike the conventional detectors working on a single-cell basis), where the first cell is associated with τ_1 and two out of $K_P - 1$ range bins are related to τ_2 and τ_3 . The remaining range bins are assumed free of signal components. The formal statement of the detection problem at hand is

$$\begin{cases} H_0 : \begin{cases} \mathbf{z}_k \sim \mathcal{CN}_N(\mathbf{0}, \mathbf{M}), & k \in \Omega_P, \\ \mathbf{r}_k \sim \mathcal{CN}_N(\mathbf{0}, \mathbf{M}), & k \in \Omega_S, \end{cases} \\ H_1 : \begin{cases} \mathbf{z}_1 \sim \mathcal{CN}_N(\alpha_1 \mathbf{v}_R, \mathbf{M}), \\ \mathbf{z}_n \sim \mathcal{CN}_N(\alpha_n \mathbf{v}_{SR}, \mathbf{M}), & n \in \Omega_P, n > 1, \\ \mathbf{z}_{2n} \sim \mathcal{CN}_N(\alpha_{2n} \mathbf{v}_S, \mathbf{M}), & 2n \leq K_P \\ \mathbf{z}_k \sim \mathcal{CN}_N(\mathbf{0}, \mathbf{M}), & k \in \Omega_P \setminus \{1, n, 2n\}, \\ \mathbf{r}_k \sim \mathcal{CN}_N(\mathbf{0}, \mathbf{M}), & k \in \Omega_S, \end{cases} \end{cases} \quad (2)$$

where the index $2n$ is due to the fact that $\tau_2 - \tau_1 = \tau_3 - \tau_2$; $\mathbf{z}_k \in \mathbb{C}^{N \times 1}$, $k \in \Omega_P = \{1, \dots, K_P\}$, are the space-time vectors (with $N = N_a N_p$ the number of space-time channels) containing the echo samples from the window under test (WUT); $\mathbf{r}_k \in \mathbb{C}^{N \times 1}$, $k \in \Omega_S = \{1, \dots, K_S\}$ with $K_S \geq N$, are the training vectors assumed free of useful signal components and statistically independent of data in WUT; $\mathbf{v}_R \in \mathbb{C}^{N \times 1}$ is the known space-time target steering vector as seen by the radar and, hence, is a function of target azimuth angle, elevation angle, and monostatic Doppler frequency (computed with respect to the radar); $\mathbf{v}_{SR} = \mathbf{v}_{S,1} + \mathbf{v}_{R,1}$ is the steering vector associated with τ_2 with $\mathbf{v}_{S,1} \in \mathbb{C}^{N \times 1}$ the steering vector associated with the path RTSR and $\mathbf{v}_{R,1} \in \mathbb{C}^{N \times 1}$ the steering vector corresponding to the path RSTR; $\mathbf{v}_S \in \mathbb{C}^{N \times 1}$ is the known space-time target signature of the RIS with respect to the radar when it reflects target echoes²³; $\alpha_1 \in \mathbb{C}$, $\alpha_n \in \mathbb{C}$, and $\alpha_{2n} \in \mathbb{C}$ are the unknown target responses, which also include channel and system effects, corresponding to the paths RTR, RTSR (RSTR), and RSTSR, respectively; and $\mathbf{M} \in \mathbb{C}^{N \times N}$ is the unknown positive definite covariance matrix of the disturbance.

III. ARCHITECTURE DESIGNS

In this section, we devise detection architectures for the above hypothesis testing problem by exploiting suitable modi-

²The steering vector is assumed known for each tested hypothesis (range-Doppler-angle cell), which in practice corresponds to a discrete grid search over the surveillance region.

³Steering vector mismatch has been well addressed in many existing works [22]–[24]; hence, for simplicity, perfect knowledge of the steering vectors is assumed herein.

fications of the GLRT design criterion leading to mathematical tractable optimization problems.

The general expression of the GLRT for problem (2) is

$$\frac{\max_{\substack{n \in \Omega_P \\ 1 < n \leq K_P/2}} \max_M \max_{\alpha} f(\mathbf{Z}_P, \mathbf{R}; \alpha, n, M, H_1)}{\max_M f(\mathbf{Z}_P, \mathbf{R}; M, H_0)} \underset{H_0}{\overset{H_1}{>}} \eta, \quad (3)$$

where η is the detection threshold whose value depends on the probability of false alarm (P_{fa}) specified by system requirements; $\mathbf{Z}_P = [\mathbf{z}_1, \dots, \mathbf{z}_{K_P}] \in \mathbb{C}^{N \times K_P}$; $\mathbf{R} = [\mathbf{r}_1, \dots, \mathbf{r}_{K_S}] \in \mathbb{C}^{N \times K_S}$; $\alpha = [\alpha_1, \alpha_n, \alpha_{2n}]^T \in \mathbb{C}^{3 \times 1}$; $f(\cdot; \cdot, H_0)$ and $f(\cdot; \cdot, H_1)$ are the joint PDFs of $\mathbf{Z}_P$ and $\mathbf{R}$ under H_0 and H_1 , respectively.

A. Estimate-and-Plug Solutions

The approach used in this subsection (also known as two-step GLRT [18]) consists in deriving the GLRT when α is known and then in replacing it with suitable estimates. The maximization under H_0 is a well-known problem (see for instance [19], [25]–[31] and references therein) and leads to

$$f(\mathbf{Z}_P, \mathbf{R}; \widehat{\mathbf{M}}, H_0) = \left[\frac{K}{\pi e \det^{1/N}(\mathbf{S}_P + \mathbf{S}_S)} \right]^{KN}, \quad (4)$$

where $\widehat{\mathbf{M}}$ is the maximum likelihood estimate of $\mathbf{M}$ under H_0 , $\mathbf{S}_P = \mathbf{Z}_P \mathbf{Z}_P^\dagger$, and $\mathbf{S}_S = \mathbf{R} \mathbf{R}^\dagger$. Given $1 < n \leq K_P/2$, the maximization with respect to $\mathbf{M}$ under H_1 leads to⁴

$$f(\mathbf{Z}_P, \mathbf{R}; \alpha, n, \widehat{\mathbf{M}}, H_1) = \left[\frac{K}{\pi e \det^{1/N}(\mathbf{Y}_{1,n,2n} + \mathbf{S}_{n,2n})} \right]^{KN}, \quad (5)$$

where $\mathbf{Y}_{1,n,2n} = (\mathbf{Z}_{1,n,2n} - \mathbf{V}\mathbf{A})(\mathbf{Z}_{1,n,2n} - \mathbf{V}\mathbf{A})^\dagger$, $\mathbf{Z}_{1,n,2n} = [\mathbf{z}_1, \mathbf{z}_n, \mathbf{z}_{2n}] \in \mathbb{C}^{N \times 3}$, $\mathbf{A} = \text{diag}(\alpha)$, $\mathbf{S}_{n,2n} = \mathbf{S}_S + \sum_{k \in \Omega_P \setminus \{1,n,2n\}} \mathbf{z}_k \mathbf{z}_k^\dagger$, and $\mathbf{V} = [\mathbf{v}_R, \mathbf{v}_{SR}, \mathbf{v}_S] \in \mathbb{C}^{N \times 3}$. Exploiting (5) and (4), the GLRT for known α can be equivalently recast as

$$\max_{\substack{n \in \Omega_P \\ 1 < n \leq K_P/2}} \frac{\det(\mathbf{S}_P + \mathbf{S}_S)}{\det[\mathbf{S}_{n,2n} + \mathbf{Y}_{1,n,2n}]} \underset{H_0}{\overset{H_1}{>}} \eta \quad (6)$$

and the adaptive GLRT is obtained by replacing α with

$$\widehat{\alpha} = \left[\frac{\mathbf{v}_R^\dagger \mathbf{S}_S^{-1} \mathbf{z}_1}{\mathbf{v}_R^\dagger \mathbf{S}_S^{-1} \mathbf{v}_R}, \frac{\mathbf{v}_{SR}^\dagger \mathbf{S}_S^{-1} \mathbf{z}_n}{\mathbf{v}_{SR}^\dagger \mathbf{S}_S^{-1} \mathbf{v}_{SR}}, \frac{\mathbf{v}_S^\dagger \mathbf{S}_S^{-1} \mathbf{z}_{2n}}{\mathbf{v}_S^\dagger \mathbf{S}_S^{-1} \mathbf{v}_S} \right]^T. \quad (7)$$

In what follows, we refer to this detector as estimate-and-plug GLRT for known α (EP-GLRT-KA).

B. Approximate GLRT

This strategy consists in minimizing an upper bound to the objective function instead of the objective function itself (under H_1). To be more definite, let us consider the following problem

$$\max_{\substack{n \in \Omega_P \\ 1 < n \leq K_P/2}} \max_{\alpha} f(\mathbf{Z}_P, \mathbf{R}; \alpha, n, \widehat{\mathbf{M}}, H_1) \quad (8)$$

⁴We omit the derivation as it is similar to that under H_0 .

and notice that the maximization with respect to α requires to solve

$$\min_{\alpha} \det[\mathbf{S}_{n,2n} + \mathbf{Y}_{1,n,2n}] \Rightarrow \min_{\alpha} \det \left[\mathbf{I} + \mathbf{X}_{1,n,2n}^\dagger \mathbf{P}_{V_S}^\perp \times \mathbf{X}_{1,n,2n} + (\mathbf{B}\mathbf{A} - \mathbf{G}_{1,n,2n})^\dagger (\mathbf{B}\mathbf{A} - \mathbf{G}_{1,n,2n}) \right], \quad (9)$$

where⁵ $\mathbf{X}_{1,n,2n} = \mathbf{S}_{n,2n}^{-1/2} \mathbf{Z}_{1,n,2n}$, $\mathbf{V}_S = \mathbf{S}_{n,2n}^{-1/2} \mathbf{V}$, $\mathbf{P}_{V_S}^\perp = \mathbf{I} - \mathbf{V}_S (\mathbf{V}_S^\dagger \mathbf{V}_S)^{-1} \mathbf{V}_S^\dagger$, $\mathbf{B} = (\mathbf{V}_S^\dagger \mathbf{V}_S)^{1/2}$, and $\mathbf{G}_{1,n,2n} = (\mathbf{V}_S^\dagger \mathbf{V}_S)^{-1/2} \mathbf{V}_S^\dagger \mathbf{X}_{1,n,2n}$. Due to the structure of $\mathbf{A}$ the above maximization is not an easy task. For this reason, we resort to another approach by considering the following problem

$$\min_{\alpha} \log \det [\mathbf{I} + \mathbf{X}_{1,n,2n}^\dagger \mathbf{P}_{V_S}^\perp \mathbf{X}_{1,n,2n} + \mathbf{D}_\alpha^{n,2n}], \quad (10)$$

where $\mathbf{D}_\alpha^{n,2n} = (\mathbf{B}\mathbf{A} - \mathbf{G}_{1,n,2n})^\dagger (\mathbf{B}\mathbf{A} - \mathbf{G}_{1,n,2n})$. In addition, instead of minimizing the last objective function, we exploit the inequality [32] $\log \det(\mathbf{X}) \leq \text{Tr}(\mathbf{X}) - N$ to write

$$\begin{aligned} \min_{\alpha} \log \det [\mathbf{I} + \mathbf{X}_{1,n,2n}^\dagger \mathbf{P}_{V_S}^\perp \mathbf{X}_{1,n,2n} + \mathbf{D}_\alpha^{n,2n}] \\ \leq \min_{\alpha} \left\{ \text{Tr} [\mathbf{I} + \mathbf{X}_{1,n,2n}^\dagger \mathbf{P}_{V_S}^\perp \mathbf{X}_{1,n,2n} + \mathbf{D}_\alpha^{n,2n}] - N \right\}. \end{aligned} \quad (11)$$

and minimize the upper bound with respect to α , namely $\min_{\alpha} \|\mathbf{B}\mathbf{A} - \mathbf{G}_{1,n,2n}\|_F^2$. As a result we obtain that

$$\tilde{\alpha} = \left[\frac{\mathbf{v}_R^\dagger \mathbf{S}_{n,2n}^{-1} \mathbf{z}_1}{\mathbf{v}_R^\dagger \mathbf{S}_{n,2n}^{-1} \mathbf{v}_R}, \frac{\mathbf{v}_{SR}^\dagger \mathbf{S}_{n,2n}^{-1} \mathbf{z}_n}{\mathbf{v}_{SR}^\dagger \mathbf{S}_{n,2n}^{-1} \mathbf{v}_{SR}}, \frac{\mathbf{v}_S^\dagger \mathbf{S}_{n,2n}^{-1} \mathbf{z}_{2n}}{\mathbf{v}_S^\dagger \mathbf{S}_{n,2n}^{-1} \mathbf{v}_S} \right]^T \quad (12)$$

Remarkably, although (7) and (12) have similar algebraic forms, they stem from different design criteria. Moreover (7) uses the secondary-data-only covariance estimate, while (12) incorporates information from non-signal cells in the primary data for joint estimation, which provides a more accurate characterization of the interference environment at the cost of higher computational complexity. The final expression of the detector is

$$\max_{\substack{n \in \Omega_P \\ 1 < n \leq K_P/2}} \frac{\det(\mathbf{S}_P + \mathbf{S}_S)}{\det[\mathbf{S}_{n,2n} + \mathbf{Q}_{n,2n}]} \underset{H_0}{\overset{H_1}{>}} \eta, \quad (13)$$

where $\mathbf{Q}_{n,m} = (\mathbf{Z}_{1,n,m} - \mathbf{V}\tilde{\mathbf{A}})(\mathbf{Z}_{1,n,m} - \mathbf{V}\tilde{\mathbf{A}})^\dagger$ and $\tilde{\mathbf{A}} = \text{diag}(\tilde{\alpha})$. The above decision scheme is referred to in the following as approximate GLRT (A-GLRT).

IV. RIS DESIGN ISSUES AND PERFORMANCE ANALYSIS

The investigated scenario comprises a radar system, the RIS, and the target in the following positions: radar location $(-30000, 200)$ m; RIS location $(0, 0)$ m; and target location $(1000, 500)$ m. Indicating by P_T and G_T the power and the gain on transmit, respectively, and by σ_{RTR} , σ_{STS} , and σ_{STR} the RCSs associated with the LOS path, RIS-RIS path, and RIS-assisted LOS, respectively, the received power for the different paths as a function of the RIS RCS are $P_R^{RTR}(\sigma_{RIS}) = \frac{P_T G_T \sigma_{RTR} A_{eff}}{(4\pi)^2 d_{RT}^4}$;

⁵Notice that it is understood that $\mathbf{V}$ is full-column rank.

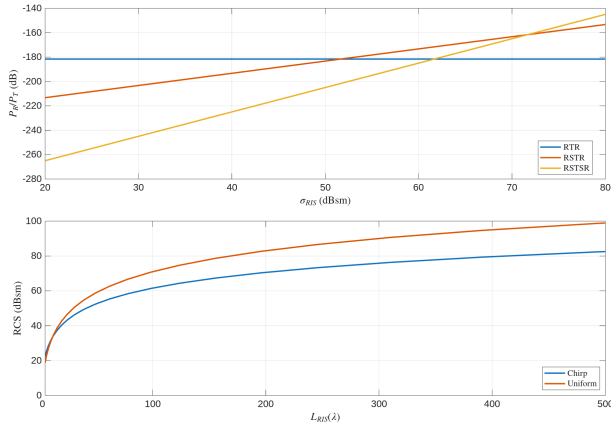


Fig. 2: Top: received power as a function of the RIS RCS for the three analyzed contributions; bottom: predicted RCS of a RIS with uniform and LFM tapering for an aperture of 10° and an operating frequency of 3 GHz.

$$P_R^{RSTR}(\sigma_{RIS}) = \frac{P_T G_T}{4\pi d_{RS}^2} \sigma_{RIS} \frac{1}{4\pi d_{ST}^2} \sigma_{STR} \frac{A_{eff}}{4\pi d_{RT}^2};$$

$$P_R^{RSTSR}(\sigma_{RIS}) = \frac{P_T G_T}{(4\pi d_{RS}^2)^2} \sigma_{RIS}^2 \frac{1}{(4\pi d_{ST}^2)^2} \sigma_{STS} A_{eff}.$$
 In Fig. 2 (on top), we plot the ratio of the received power over transmitted power as a function of the RIS RCS σ_{RIS} , which sweeps from 20 dBsm to 80 dBsm. In order to compute the size of the RIS, the RIS is tapered through linear frequency modulated (LFM) phase only weights along the x -axis, while maintaining uniform illumination along y . At boresight, the RCS of the LFM-tapered aperture can be asymptotically estimated using the stationary-phase method as $\sigma_{chirp}^\infty \approx \frac{8\pi}{\lambda^2} \frac{L_{RIS}^2}{K_x}$, where K_x is the chirp rate, λ is the operating wavelength, L_{RIS} is the length of a RIS side, E_0 is the electric field, and ζ_0 is the free space impedance. The comparison between the LFM tapering and the uniform tapering (that is a benchmark) is reported in Fig. 2 (on bottom).

Now, we focus on the performance assessment of the considered detectors. To this end, the performance metric is the probability of detection (P_d) for a preassigned value of the P_{fa} as a function of the SINR for the monostatic path defined as $\text{SINR}_{RTR} = |\alpha_1|^2 \mathbf{v}_R \mathbf{M}^{-1} \mathbf{v}_R$. The above probabilities are estimated by resorting to 10^4 and $100/P_{fa}$ Monte Carlo trials for the P_d and P_{fa} , respectively. The radar system is equipped with a uniform linear array formed by N_a vertically aligned sensors, whose inter-element spacing is equal to $\lambda/2$. Based on the considered scenario, the angles of arrival are $\theta = 0.5^\circ$ for $\mathbf{v}_R$ and $\theta = -0.4^\circ$ for $\mathbf{v}_S$; moreover, since only spatial channels are considered in the simulation setup ($N = N_a$), it follows that $\mathbf{v}_{S,1} = \mathbf{v}_S$ and $\mathbf{v}_{R,1} = \mathbf{v}_R$.

The RIS size is such that the bistatic paths generate echoes with powers that are about 5, 10, 15 dB stronger than those from the monostatic path. Thus, once α_1 is set, we consider $\alpha_n = \alpha_{2n} \approx 2\alpha_1, 3\alpha_1, 6\alpha_1$. Moreover, the range bins containing the returns from the bistatic paths are indexed by $n = 3$ for the path RTR (or RSTR) and $2n = 6$ for the path RSTSR. With these values in mind, we set $K_P = 6$. As for the interference component, it is modeled as a complex Gaussian

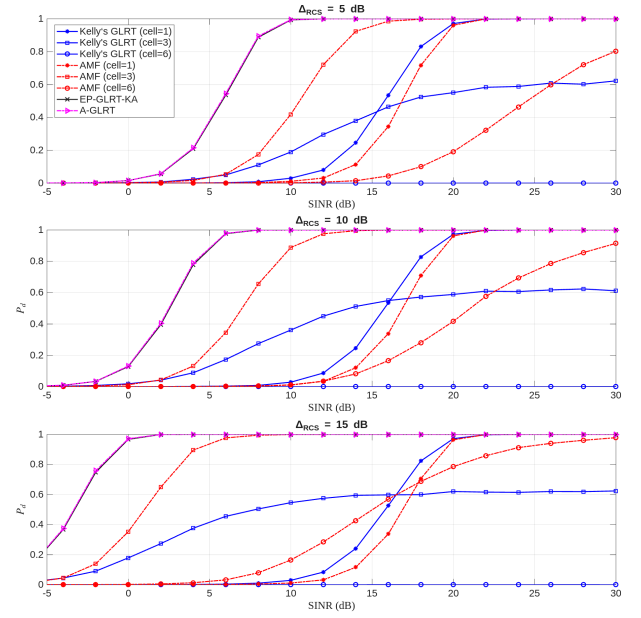


Fig. 3: P_d versus SINR for the considered detectors and different powers between the echoes from the direct and multistatic paths.

random vector with covariance matrix $\mathbf{M} = \sigma_n^2 \mathbf{I} + \mathbf{M}_c$, where $\sigma_n^2 = 1$ is the noise power and $\mathbf{M}_c$ is the clutter covariance matrix whose (n, m) th entry is $\sigma_c^2 \rho^{|n-m|}$ with $\rho = 0.9$ the one-lag correlation coefficient and $\sigma_c^2 > 0$ the clutter power set according to the clutter-to-noise ratio (CNR) defined as $\text{CNR} = 10 \log_{10}(\sigma_c^2 / \sigma_n^2) = 25$ dB. Finally, in the following numerical examples, we assume $N = 16$, $K = 24$, and $P_{fa} = 10^{-4}$.

In Fig. 3, we plot P_d versus SINR for different values of RCS difference between direct and bistatic paths (Δ_{RCS}) in comparison with the conventional detectors Kelly's GLRT [25] and the adaptive matched filter (AMF) [18], each considered for the direct-path-only scenario (cell=1) and for the RIS-assisted bins (cell=3,6). The figure points out that the proposed detectors can overcome all the considered conventional detectors, whose performance are also affected by the steering mismatch, also when fed by the range bins containing the bistatic echoes. In addition, it is important to observe that when the conventional detectors declare the presence of a target in these range bins, the information related to target position is not correct unlike the proposed architectures.

V. CONCLUSIONS

In this paper, we have proposed a RIS-assisted radar system capable of capitalizing the energy reflected by suitably shaped stealth targets. In fact, such energy is redirected by edges and surfaces of this kind of targets into irrelevant directions and the role of the RIS consists in intercepting it. In this context, we have developed two adaptive detectors capable of exploiting the echoes from the RIS also correctly locating the target position. Remarkably, this approach is significantly more effective than conventional detectors.

REFERENCES

- [1] E. Björnson, O. Özdogan, and E. G. Larsson, "Reconfigurable Intelligent Surfaces: Three Myths and Two Critical Questions," *IEEE Communications Magazine*, vol. 58, no. 12, pp. 90–96, 2020.
- [2] C. Pan, H. Ren, K. Wang, J. F. Kolb, M. Elkhachan, M. Chen, M. Di Renzo, Y. Hao, J. Wang, A. L. Swindlehurst, X. You, and L. Hanzo, "Reconfigurable Intelligent Surfaces for 6G Systems: Principles, Applications, and Research Directions," *IEEE Communications Magazine*, vol. 59, no. 6, pp. 14–20, 2021.
- [3] Z. Fu, X. Zou, Y. Liao, G. Lai, Y. Li, and K. L. Chung, "A brief review and comparison between transmitarray antennas, reflectarray antennas and reconfigurable intelligent surfaces," in *2022 IEEE Conference on Telecommunications, Optics and Computer Science (TOCS)*. IEEE, 2022, pp. 1192–1196.
- [4] F. Costa and M. Borgese, "Electromagnetic model of reflective intelligent surfaces," *IEEE Open Journal of the Communications Society*, vol. 2, pp. 1577–1589, 2021.
- [5] D. F. Sievenpiper, J. H. Schaffner, H. J. Song, R. Y. Loo, and G. Tantonan, "Two-dimensional beam steering using an electrically tunable impedance surface," *IEEE Transactions on antennas and propagation*, vol. 51, no. 10, pp. 2713–2722, 2003.
- [6] I. Alamzadeh, G. C. Alexandropoulos, N. Shlezinger, and M. F. Imani, "A reconfigurable intelligent surface with integrated sensing capability," *Scientific reports*, vol. 11, no. 1, p. 20737, 2021.
- [7] L. Wu, K. Lou, J. Ke, J. Liang, Z. Luo, J. Y. Dai, Q. Cheng, and T. J. Cui, "A wideband amplifying reconfigurable intelligent surface," *IEEE Transactions on Antennas and Propagation*, vol. 70, no. 11, pp. 10623–10631, 2022.
- [8] F. Liu, C. Masouros, A. P. Petropulu, H. Griffiths, and L. Hanzo, "Joint Radar and Communication Design: Applications, State-of-the-Art, and the Road Ahead," *IEEE Transactions on Communications*, vol. 68, no. 6, pp. 3834–3862, 2020.
- [9] D. Li, B. Tang, X. Wang, W. Wu, and L. Xue, "Codesign for Spectral Coexistence Between RIS-Aided MIMO Radar and MIMO Communication Systems," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 60, no. 6, pp. 8166–8183, 2024.
- [10] Y. He, Y. Cai, H. Mao, and G. Yu, "RIS-Assisted Communication Radar Coexistence: Joint Beamforming Design and Analysis," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 7, pp. 2131–2145, 2022.
- [11] X. Wang, Z. Fei, J. Guo, Z. Zheng, and B. Li, "RIS-Assisted Spectrum Sharing Between MIMO Radar and MU-MISO Communication Systems," *IEEE Wireless Communications Letters*, vol. 10, no. 3, pp. 594–598, 2021.
- [12] Z. Liu, H. Zhang, T. Huang, F. Xu, and Y. C. Eldar, "Hybrid RIS-Assisted MIMO Dual-Function Radar-Communication System," *IEEE Transactions on Signal Processing*, vol. 72, pp. 1650–1665, 2024.
- [13] G. Mylonopoulos, L. Venturino, E. Grossi, S. Buzzi, and C. D'Elia, "Integrated Communication and RIS-Aided Track-Before-Detect Radar Sensing," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 4519–4532, 2025.
- [14] S. Tonissen and R. Evans, "Performance of dynamic programming techniques for Track-Before-Detect," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 32, no. 4, pp. 1440–1451, 1996.
- [15] S. Tonissen and Y. Bar-Shalom, "Maximum likelihood track-before-detect with fluctuating target amplitude," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 34, no. 3, pp. 796–809, 1998.
- [16] A. Elgamal, W. Alshrafi, T. Dallmann, and P. Knott, "Enhancing polarization diversity in RIS-aided Integrated Communication and Sensing Networks," in *2025 22nd European Radar Conference (EuRAD)*, 2025, pp. 307–310.
- [17] "Low Observable Principles, Stealth Aircraft and Anti-Stealth Technologies," *Journal of Computations & Modelling*, vol. 4, no. 8, pp. 129–165, 2014.
- [18] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 28, no. 1, pp. 208–216, 1992.
- [19] D. Orlando, G. Ricci, and L. L. Scharf, "A Unified Theory of Adaptive Subspace Detection Part I: Detector Designs," *IEEE Transactions on Signal Processing*, vol. 70, pp. 4925–4938, 2022.
- [20] P. Addabbo, D. Orlando, G. Ricci, and L. L. Scharf, "A Unified Theory of Adaptive Subspace Detection Part II: Numerical Examples," *IEEE Transactions on Signal Processing*, vol. 70, pp. 4939–4950, 2022.
- [21] M. A. Richards, J. A. Scheer, and W. A. Holm, *Principles of Modern Radar: Basic Principles*. Raleigh, NC: Scitech Publishing, 2010.
- [22] Y. Jin, C. Yin, T. Wang, C. Hao, and D. Orlando, "An adaptive target detection architecture for mismatched signals," *IEEE Signal Processing Letters*, vol. 32, pp. 1860–1864, 2025.
- [23] W. Liu, W. Xie, R. Li, F. Gao, X. Hu, and Y. Wang, "Adaptive detection in the presence of signal mismatch," *Journal of Systems Engineering and Electronics*, vol. 26, no. 1, pp. 38–43, 2015.
- [24] J. Liu, W. Liu, B. Chen, H. Liu, and H. Li, "Detection probability of a cfar matched filter with signal steering vector errors," *IEEE Signal Processing Letters*, vol. 22, no. 12, pp. 2474–2478, 2015.
- [25] E. J. Kelly, "An adaptive detection algorithm," *IEEE Transactions on Aerospace and Electronic Systems*, no. 2, pp. 115–127, 1986.
- [26] E. Conte, A. De Maio, and G. Ricci, "GLRT-based adaptive detection algorithms for range-spread targets," *IEEE Transactions on Signal Processing*, vol. 49, no. 7, pp. 1336–1348, July 2001.
- [27] W. Liu, J. Liu, C. Hao, Y. Gao, and Y.-L. Wang, "Multichannel adaptive signal detection: basic theory and literature review," *Science China Information Sciences*, vol. 65, no. 2, p. 121301, 2022.
- [28] F. Bandiera, D. Orlando, and G. Ricci, *Advanced Radar Detection Schemes Under Mismatched Signal Models*. Switzerland AG: Springer Cham, 2009.
- [29] W. Liu, W. Xie, J. Liu, and Y. Wang, "Adaptive Double Subspace Signal Detection in Gaussian Background—Part I: Homogeneous Environments," *IEEE Transactions on Signal Processing*, vol. 62, no. 9, pp. 2345–2357, 2014.
- [30] O. Besson, A. Coluccia, E. Chaumette, G. Ricci, and F. Vincent, "Generalized Likelihood Ratio Test for Detection of Gaussian Rank-One Signals in Gaussian Noise With Unknown Statistics," *IEEE Transactions on Signal Processing*, vol. 65, no. 4, pp. 1082–1092, 2017.
- [31] Y. Jin and B. Friedlander, "A CFAR adaptive subspace detector for second-order Gaussian signals," *IEEE Transactions on Signal Processing*, vol. 53, no. 3, pp. 871–884, 2005.
- [32] H. Lütkepohl, *Handbook of Matrices*. Wiley, 1997.